QUESTIONS FOR CHAPTER 13—MONITORING AND REPORTING
问题：第十三章—监控与报告

1. You are the **project manager** for a project and have worked with your team members to establish the following criteria for **tracking the work** of the project. Develop the **work/time relationship curve** for the project similar to **Figure 13-7** in your book.

   你是该项目的**项目经理**，并已与团队成员共同确定了用于**跟踪项目工作**的以下标准。请绘制该项目的**工作/时间关系曲线**，类似于书中的**图 13-7**。

Task	Weight multiplier	Project timing
Review backup material	0.05	0%–10%
Preliminary calculations	0.10	10%–25%
Initial drafting	0.25	15%–45%
Final calculations	0.20	35%–60%
Production drawings	0.30	50%–90%
Drawing approvals	0.10	90%–100%

FIGURE 13-7 Work/time relationship for all design work (E4th Ch13 P477)

Relationships between Time and Work

时间与工作的关系

Measurement of **design work** is difficult because design is a **creative process** that involves **ideas, calculations, evaluation of alternatives**, and other tasks that are not **physically measurable quantities**. Considerable **time and cost** can be expended in performing these tasks before end results such as **drawings, specifications, reports**, etc., which are **measurable quantities of work**, are ever seen.

设计工作的衡量是困难的，因为设计是一个**创造性过程**，涉及**创意、计算、替代方案的评估**以及其他**无法量化的任务**。在产生**图纸、规范、报告**等**可量化的工作成果**之前，可能需要花费大量的**时间和成本**来执行这些任务。

The measurement of design is further complicated because of the **diversity of work**. For example, all the **design calculations** may be complete, half of the **drawings** may be produced, and yet only one-fourth of the **specifications** may be written. For this situation, it is difficult to determine how much work has been accomplished because the work that is produced does not have a **common unit of measure**. Because of this, a **percentage of completion** is commonly used as a **unit of measure** of **design work**. The **criteria** for determining the **percent complete** for measuring work must be developed and confirmed **in writing** with the **project team members** prior to commencing design. This provides a **common basis** for the **monthly evaluation** of progress.

设计的衡量因**工作的多样性**而更加复杂。例如，所有的**设计计算**可能已经完成，半数的**图纸**已经绘制完成，但仅有四分之一的**规范**完成了编写。在这种情况下，很难确定已完成的工作量，因为所产生的工作**没有共同的计量单位**。因此，通常使用**完成百分比**作为**设计工作的计量单位**。在开始设计之前，必须与**项目团队成员**共同制定并**书面确认**用于衡量工作的**完成百分比标准**，这为**每月进展评估**提供了**共同的依据**。

A **weighting multiplier** can be assigned to **design tasks** to define the **magnitude of effort** that is required to achieve each task. The sum of the **weighting factors** is **1.0**, which represents **100% of the total design effort**. The determination of each **weight** should be a joint effort between the **project manager** and the **designer** who is responsible for performing the work. This should be done **before starting work**.

可以为**设计任务**分配一个**权重乘数**，以定义完成每项任务所需的**工作量**。**权重系数**之和为 **1.0**，表示 **100%的设计工作总量**。每个**权重的确定**应由**项目经理**和负责执行工作的**设计人员**共同完成。这应在**工作开始之前**完成。

A significant amount of **overlap work** is necessary during the **design process**. For example, **initial drafting** normally starts before the **design calculations** are finished. Likewise, **final calculations** are often not complete before the **final production drawings** are started. The **project manager** and their **team members** can jointly define the **overlap of related work** to show the **timing of tasks** throughout the duration of a project. **Table 13-5** is an illustrative example that lists **work items, weight multiplier**, and **estimated timing** for each task required in a **design effort**. Further division of **weight multipliers** within each category may also be necessary.

在**设计过程中**，需要大量的**工作重叠**。例如，**初步绘图**通常在**设计计算**完成之前开始。同样，**最终计算**通常在**最终图纸**绘制之前尚未完成。**项目经理**及其**团队成员**可以共同定义**相关工作的重叠**，以显示项目过程中**任务的时间安排**。**表 13-5** 是一个**示例**，列出了**设计工作所需的工作项目、权重乘数**以及**每项任务的估计时间**。在每个类别中进一步划分**权重乘数**可能也是必要的。

The information in **Table 13-5** is provided for illustrative purposes only. Because each **project is unique**, it is necessary to define the **weight multipliers** that are appropriate for each **individual project**. The **timing of design work** depends on the **availability of personnel**. This information can be compiled from a **summary of the individual design work packages**.

表 13-5 中的信息仅供示例参考。由于每个**项目都是独特的**，因此需要为每个**单独的项目**定义合适的**权重乘数**。**设计工作的时间安排**取决于**人员的可用性**。此信息可以从**各个设计工作包的汇总**中编制。

TABLE 13-5 Illustrative Weight Multipliers for Design Work
表 13-5 设计工作权重乘数示例

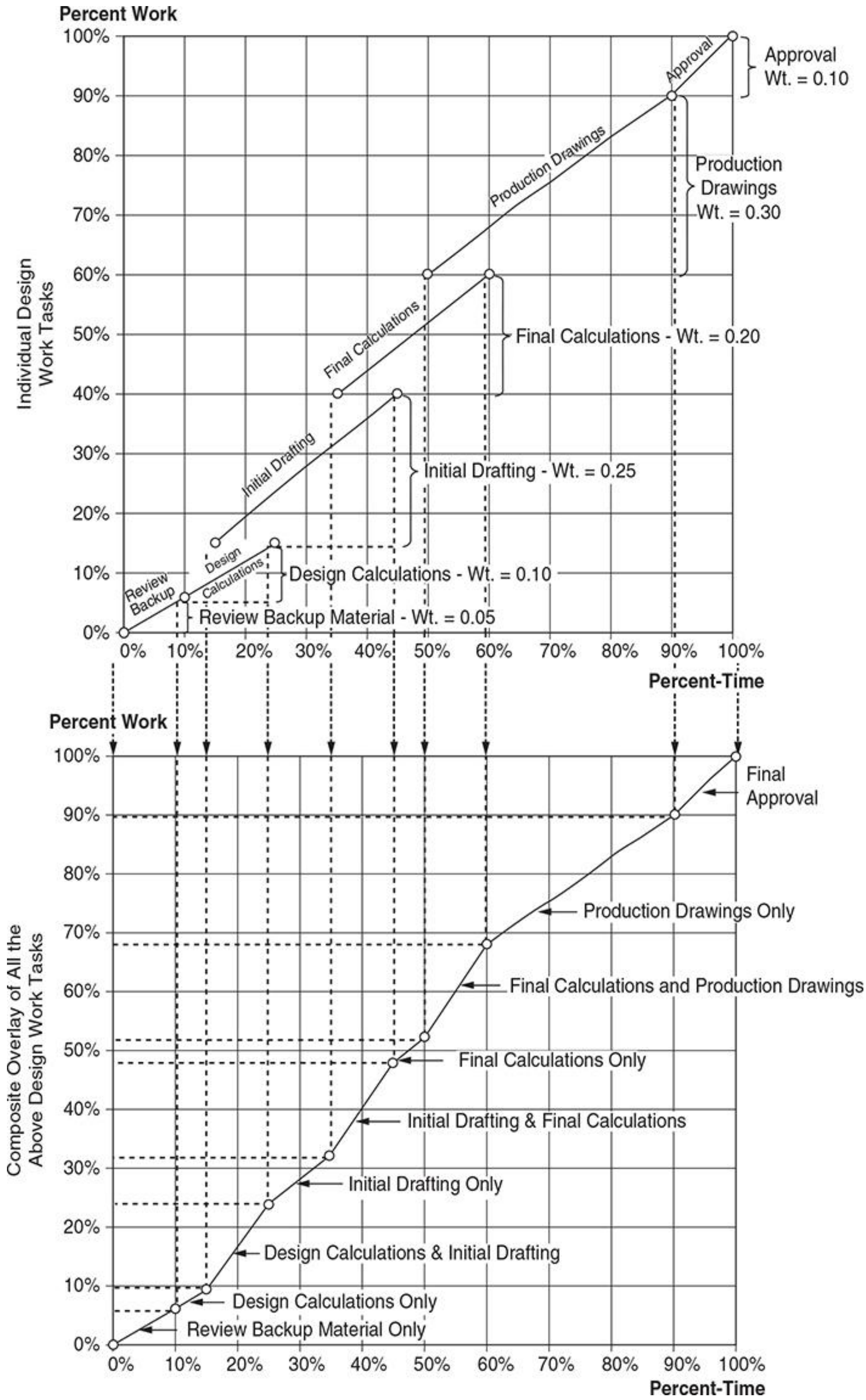
Design Work	Weight Multiplier	Project Timing
Review backup material	0.05	0%–10%
Design calculations	0.10	10%–25%
Initial drafting	0.25	15%–45%
Final calculations	0.20	35%–60%
Production drawings	0.30	50%–90%
Drawing approval	0.10	90%–100%
	1.00	

To manage the **overall design effort**, a **work/time curve** can be developed from the information in **Table 13-5**, see **Figure 13-7**. The **upper portion** of **Figure 13-7** is a series of **graphs** that represent each **design task individually**, arranged in the **order of occurrence**. The **slope** of each graph is the **ratio of the weight multiplier** to the **time required for the work** to be performed. The **lower portion** of **Figure 13-7** is the **work/time curve** for the **entire design effort** and is obtained by a **composite, superposition** of the individual graphs. This curve represents the **planned accomplishment of work** and serves as a **basis of control** for comparison of **actual work accomplished**. It can be **superimposed** onto the **time/cost S-curve** that was discussed in the preceding section. This forms the **overall integrated cost/schedule/work curve** that is discussed later in subsequent sections of this chapter.

为了管理**整体设计工作**，可以根据**表 13-5** 中的信息绘制一个**工作/时间曲线**，见**图 13-7**。**图 13-7 的上半部分**是一系列**图表**，分别代表各个**设计任务**，按**发生顺序排列**。每个图表的**斜率**是**权重乘数与完成该工作所需时间的比率**。**图 13-7 的下半部分**是**整个设计工作的工作/时间曲线**，通过各个图表的**组合叠加**得到。此曲线表示**计划完成的工作**，并作为**控制的基础**，用于比较**实际完成的工作**。它可以与上一节讨论的**时间/成本 S 曲线**叠加。这形成了**总体的成本/进度/工作集成曲线**，将在本章后续部分讨论。

FIGURE 13-7 Work/time relationship for all design work

图 13-7 所有设计工作的时间/工作关系



2. The **design team** for a project has met and determined the **grouping of tasks** required for the design effort into **six categories**. The team has also agreed on the following **criteria for weight multipliers** to be used to track the design effort. Prepare a **graph** that shows the **relationship between work and time**.

   该项目的**设计团队**已经开会并确定了设计工作所需的任务**分组为六个类别**。团队还就**跟踪设计工作**所使用的**权重乘数标准**达成了一致。请绘制一个**图表，**显示**工作与时间之间的关系**。

Design work	Weight multiplier	Project timing
Initial configurations	0.15	0%–10%
Initial design calculations	0.10	5%–15%
Preliminary layouts	0.05	10%–25%
Final calculations	0.25	20%–50%
CADD drawings	0.40	45%–90%
Design approval	0.05	90%–100%

3. At **50% into the project duration**, the status report for the project in **Question 2** shows the following information. Prepare a **graph** that shows the **time relationship of the planned work and actual work**. Is the project **ahead of or behind schedule**?

在**项目时间过半（50%）时，问题 2 中的项目状态报告显示了以下信息。请绘制一个图表，**展示**计划工作与实际工作在时间上的关系**。该项目是**进度超前**还是**落后**？

Design work	% Complete
Initial configurations	100%
Initial design calculations	100%
Preliminary layouts	80%
Final calculations	70%
CADD drawings	10%
Design approval	0%

4. The following data represents the **anticipated cost** and **planned work** during the **design phase**. Prepare an **integrated planned cost/time/work graph** for the project, similar to **Figure 13-9** in your book.

以下数据表示**设计阶段**中的**预期成本**和**计划工作**。请为该项目绘制一个**集成的计划成本/时间/工作图表**，类似于书中的**图 13-9**。

Expected time	Anticipated cost	Planned work
0%	0%	0%
10%	5%	5%
20%	10%	12%
30%	12%	20%
40%	30%	25%
50%	52%	30%
60%	80%	45%
70%	84%	62%
80%	92%	85%
90%	96%	95%
100%	100%	100%

也可以写作（2nd Ch9Q4）：

The following data represent the information that has been **jointly compiled** by the **design team** for a project. These data represent the **planned cost** and the **work anticipated** during the **design phase**. Prepare an **integrated planned cost/time/work graph** for the project.

以下数据表示**设计团队**为该项目**共同汇编**的信息。这些数据代表了**设计阶段**的**计划成本**和**预期工作量**。请为该项目绘制一个**集成的计划成本/时间/工作图表**。

FIGURE 13-9 Integrated cost/schedule/work graph (E4th Ch13 P485)

Integrated Cost/Schedule/Work

综合成本/进度/工作

Experienced **project managers** are familiar with the problems of using only **partial information**, such as only **costs** or **time** to track the status of a project. To illustrate, half of a **project budget** may be expended by the midpoint of the scheduled duration, but only **20% of the work** may be accomplished. Monitoring only **time and cost** would indicate the project is going well; however, upon completion of the project, there would likely be a **cost overrun** and a **delay in schedule** because the **measurement of work** was not included in the **project control system**. Thus, a project manager must develop an **integrated cost/schedule/work system** which provides **meaningful feedback** during the project rather than afterward. The **status of a project** can then be determined, and **prompt corrective actions** taken to minimize the **cost impacts** to the project.

有经验的项目经理了解仅使用**部分信息**（例如仅考虑**成本或时间**）来跟踪项目状态的问题。举例来说，项目预算的一半可能会在**计划时间过半**时消耗完，但可能只完成了 **20%的工作**。仅监控**时间和成本**会让人觉得项目进展顺利；然而，在项目完成时，可能会发现**成本超支**和**进度延误**，因为**工作量的衡量**未被纳入**项目控制系统**。因此，项目经理必须开发一个**集成的成本/进度/工作系统**，以便在项目进行过程中提供**有意义的反馈**，而不是事后反馈。这样可以确定**项目的状态**，并及时采取**纠正措施**，以尽量减少项目的**成本影响**。

The preceding sections presented the relationships of **cost/time** and **work/time** for **project control**. However, evaluating these relationships separately does not provide an **accurate status** of a project. A **cost/schedule/work graph** can be prepared that shows the **integrated relationship** of the three basic components of a project: **scope (work), budget (cost)**, and **schedule (time)**. **Figure 13-9** is a graph that links **costs** on the **left-hand ordinate**, **time** on the **abscissa**, and **work** on the **right-hand ordinate**. The upper curve is simply the **cost/time S-curve** that has been discussed in previous sections of this book. The **lower work/time curve** shows the **relationship between work and time** throughout the duration of the project. Thus, the graph is simply a **composite overlay** of the information previously presented.

在之前的章节中介绍了**成本/时间**和**工作/时间**的关系，用于**项目控制**。然而，分别评估这些关系并不能准确反映**项目的状态**。可以制作一个**成本/进度/工作图表**，显示项目的三个基本组成部分之间的**集成关系：范围（工作）、预算（成本）和进度（时间）**。**图 13-9** 是一张将**成本**标在**左侧纵坐标**、**时间**标在**横坐标**、**工作**标在**右侧纵坐标**的图表。上方的曲线只是之前章节讨论过的**成本/时间 S 曲线**。下方的**工作/时间曲线**显示了**工作与时间的关系**，贯穿项目的整个过程。因此，该图表只是之前所介绍信息的**组合叠加**。

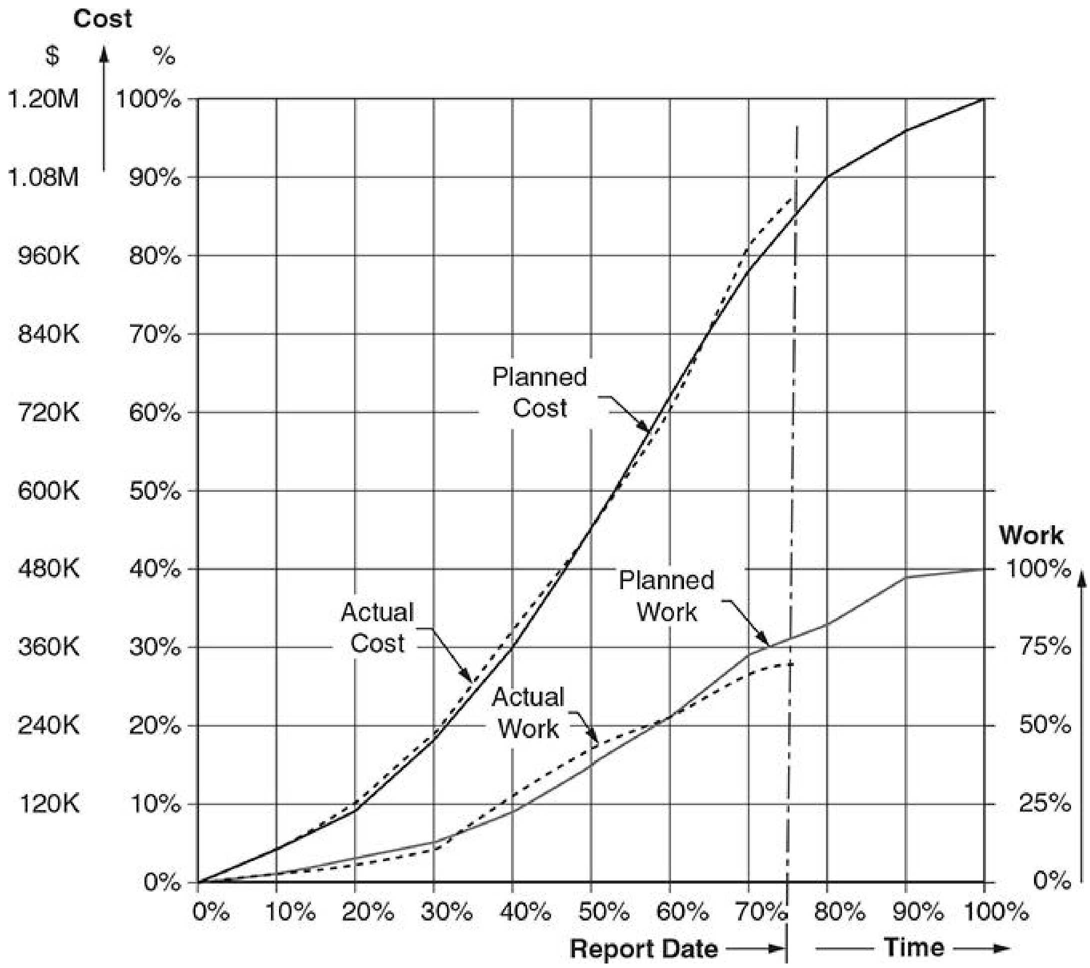


FIGURE 13-9 Integrated cost/schedule/work graph.

图 **13-9** 综合成本**/**进度**/**工作图表。

5. The data shown below represents the **planned time, cost, and work** at the beginning of a project, similar to **Example 13-1** in your book. Prepare the **integrated time/cost/work graph**.

以下数据显示了项目开始时的**计划时间、成本和工作**，类似于书中的**示例 13-1**。请绘制**集成的时间/成本/工作图表**。

Time	Cost	Work
0%	0%	0%
10%	5%	5%
20%	15%	10%
30%	20%	20%
40%	30%	30%
50%	45%	40%
60%	65%	50%
70%	80%	70%
80%	90%	80%
90%	95%	90%
100%	100%	100%

Evaluate the **status report** for the project for the following **three scenarios**, assuming the **status report is issued at 70% of the project duration**. For each of the three scenarios, determine if the project is **behind or ahead of schedule**. Is there a **cost overrun or underrun**, and if so, by how much?

根据项目的以下**三种情况**的**状态报告**评估项目状态，假设**状态报告在项目时间的 70%时发布**。在每种情况中确定项目是**进度超前**还是**落后**。是否存在**成本超支**或**节省**，如果有，是多少？

a. Reported cost to date is **75%** and work accomplished is reported as **50%**.
b. Reported cost to date is **95%** and work accomplished is reported as **70%**.
c. Reported cost to date is **85%** and work accomplished is reported as **80%**.

a. 截至目前的报告成本为 **75%**，已完成工作为 **50%**。
b. 截至目前的报告成本为 **95%**，已完成工作为 **70%**。
c. 截至目前的报告成本为 **85%**，已完成工作为 **80%**。

补充问题 (2nd Ch9Q5)：

d. Reported cost to date is 55% and work accomplished is reported as 40%

E4th Ch13 Example 13-1

The following data represent the information that has been **jointly compiled** by the various **disciplines of the design team** for a project. These data represent the **planned cost and work** anticipated during the **design phase**. Prepare an **integrated planned cost/time/work graph** for the project.

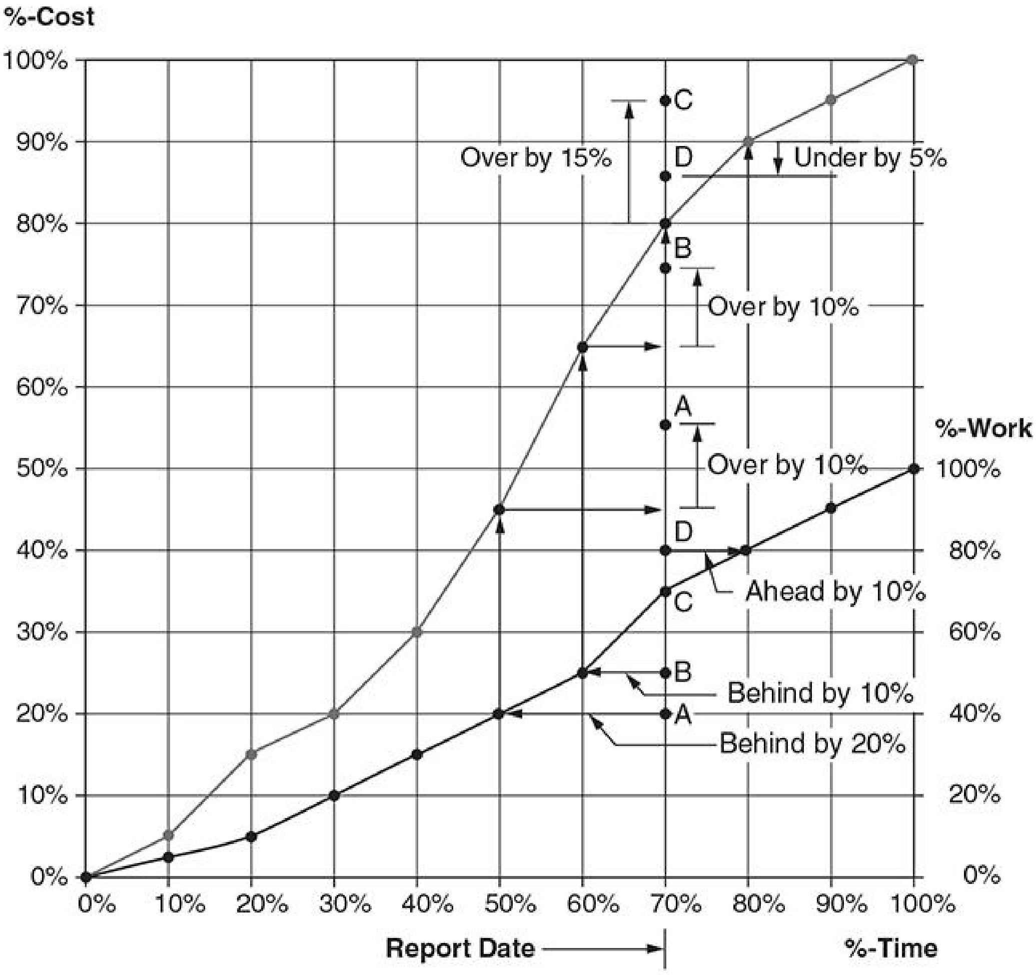
以下数据表示**设计团队各个学科**共同汇编的信息。这些数据代表了**设计阶段**中的**计划成本和预期工作**。请为该项目绘制一个**集成的计划成本/时间/工作图表**。

Expected Time	Anticipated Cost	Planned Work
0%	0%	0%
10%	5%	5%
20%	15%	10%
30%	20%	20%
40%	30%	30%
50%	45%	40%
60%	65%	50%
70%	80%	70%
80%	90%	80%
90%	95%	90%
100%	100%	100%

Evaluate the status report for the project for the following **scenarios**, assuming the **status report** is issued at **70% of the project duration**. For each scenario, determine if the project is **behind or ahead of schedule**. Is there a **cost overrun or underrun**, and if so, by **how much**? A **graphical evaluation** of the **four scenarios** is shown below, followed by a **summary table**.

根据项目的**状态报告**评估以下**情境**，假设**状态报告**在**项目时间的 70%时发布。对于每种情境，确定项目是否进度滞后**或**超前**。是否存在**成本超支**或**节省**，如果有，是多少？下方展示了**四种情境的图形评估**，随后是一个**汇总表**。

Scenario	Status report at 70% of the project duration
A	Reported cost to date is 55% and work accomplished is reported as 40%
B	Reported cost to date is 75% and work accomplished is reported as 50%
C	Reported cost to date is 95% and work accomplished is reported as 70%
D	Reported cost to date is 85% and work accomplished is reported as 80%



Scenario	Schedule	Cost
A	Behind by 20%	Over by 10%
B	Behind by 10%	Over by 10%
C	On schedule	Over by 15%
D	Ahead by 10%	Under by 5%

6. The following **weight multipliers** have been established for the **construction phase** of a project. Prepare a **graph** that shows the **relationship between accomplished work and time**, similar to **Figure 13-8** in your book.

已为项目的**施工阶段**确定了以下**权重乘数**。请绘制一个**图表**，显示**完成的工作与时间之间的关系**，类似于书中的**图 13-8**。

Facility	Weight multiplier	Project timing
Site-work	0.10	0%–15%
Wood frame building	0.25	10%–25%
Concrete building	0.35	15%–40%
Metal building	0.20	25%–80%
Landscape	0.10	75%–100%

FIGURE 13-8 Work/time relationship for construction of the entire project (E4th Ch13 P483)

Relationships between Time and Work - Continued

时间与工作的关系

Construction involves many types of different work that have different **units of measure**, such as **cubic yards, square feet, pounds**, or **each**. Thus, it is convenient to use **percentage** as a **unit of measure** for **management and control** of overall construction.

施工包含多种不同类型的工作，这些工作有不同的**计量单位**，例如**立方码、平方英尺、磅**或**件**。因此，使用**百分比**作为**整体施工的管理和控制单位**较为方便。

The **procedure** used for **measurement of design** can also be applied to **construction**. For example, a project may consist of three major facilities: **site work, a concrete office building**, and a **pre-engineered metal building**. A **weight multiplier** can be applied to each of the three major facilities, along with their **planned sequence of occurrence** (reference **Table 13-6**). **Table 13-6** only lists three major parts of the project. A more **accurate definition of the planned work** can be achieved by **dividing each of the major facilities into smaller components**. For example, **site work** can be divided into **grading, drainage, paving, landscaping**, etc. Likewise, each of the two buildings can be divided into smaller components. Regardless of the **level of detail of the project**, the **sum of all the weight multipliers** would be equal to **1.0**, representing **100% of the project**. The **weight and timing** of each major facility is established by **key participants of the project team**, before starting construction, to serve as a **basis of control** during construction.

用于**设计衡量**的**方法**也可以应用于**施工**。例如，一个项目可能由三个主要设施组成：**场地工程、一栋混凝土办公楼**和一栋**预制金属建筑**。可以为这三个主要设施中的每一个应用一个**权重乘数**，并按照它们的**计划顺序**（参见**表 13-6**）。**表 13-6** 仅列出了项目的三个主要部分。通过**将每个主要设施进一步划分为更小的组成部分**，可以更**准确地定义计划的工作**。例如，**场地工程**可以细分为**平整、排水、铺路、景观美化**等。同样，这两栋建筑中的每一栋也可以分成更小的组成部分。无论**项目的细节级别**如何，**所有权重乘数之和**都应等于 **1.0**，表示**项目的 100%**。**每个主要设施的权重和时间**由**项目团队的关键参与者**在施工开始前共同确定，作为**施工期间的控制基础**。

TABLE 13-6 Illustrative Weight Multiplier for Construction

表 13-6 施工权重乘数示例

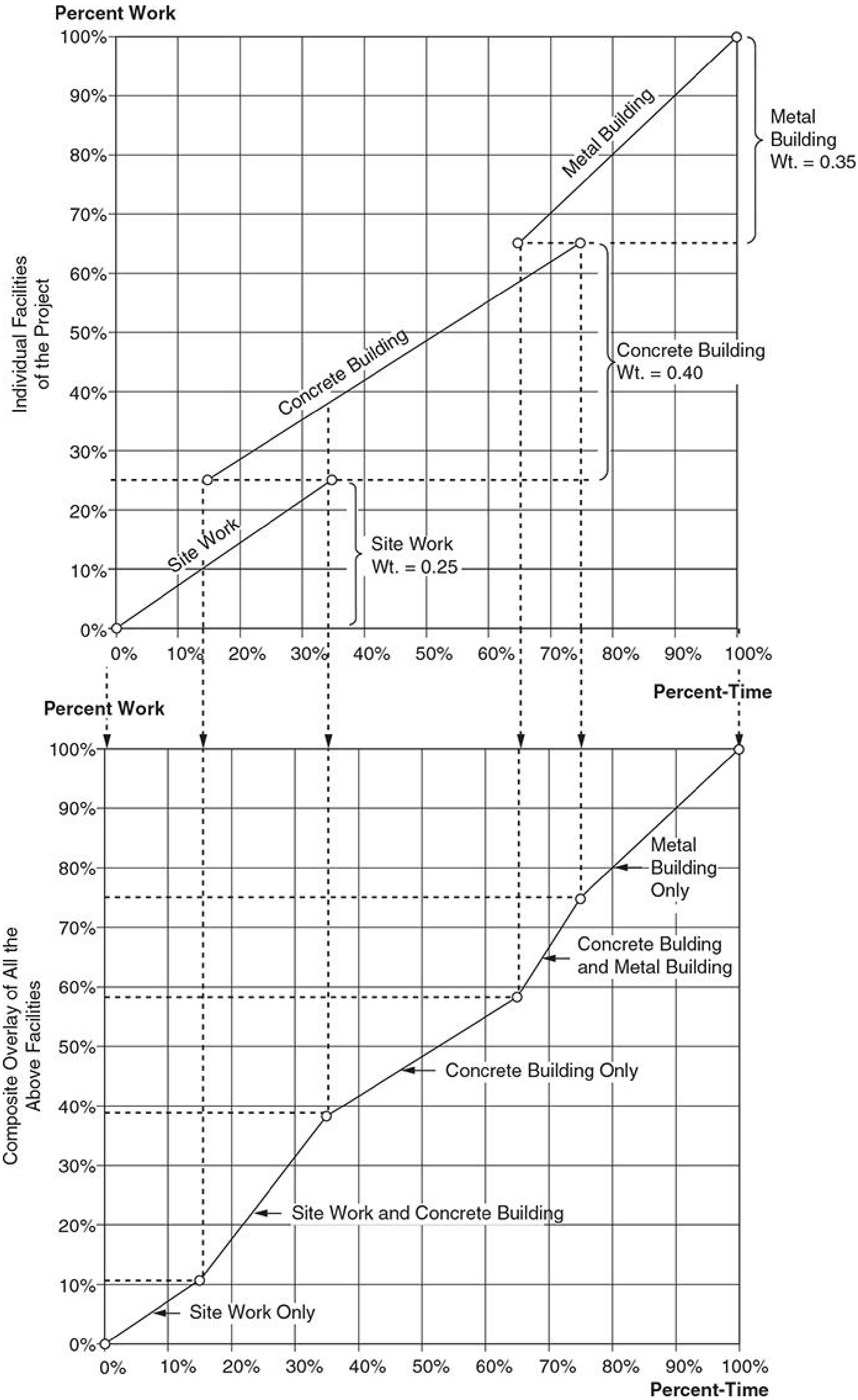
Facility	Weight Multiplier	Project Timing
Site work	0.25	0%–35%
Concrete building	0.40	15%–75%
Metal building	0.35	65%–100%
	1.00	

Figure 13-8 shows the **composite of the planned accomplishment of work**. The **upper portion** of **Figure 13-8** is a graphical display of the **overlap and sequence** of each of the three **major facilities**. The **lower portion** is the **integrated work/time curve** for the **entire project** and is obtained by a **composite superposition** of the three individual graphs of each major facility.

图 13-8 展示了**计划完成工作的综合情况**。**图 13-8 的上半部分**是各**主要设施的重叠和顺序**的图形显示。**下半部分**是整个项目的**集成工作/时间曲线**，通过每个主要设施的**单独图表的组合叠加**得到。

FIGURE 13-8 Work/time relationship for construction of the entire project.

图 13-8 整个项目施工的时间/工作关系



The **procedure** is a **top-level summary** of all facilities in the **total project**. This same procedure can be used for **each facility**, or **parts of a facility**, depending on the **complexity of the project** and the **level of control** that is desired by the **project manager**.

该**方法**是对**整个项目**中所有设施的**顶层汇总**。根据**项目的复杂性**和**项目经理所需的控制级别**，同样的程序可以用于**每个设施**或**设施的某些部分**。

At the **lowest level**, it may be possible to use a **unit of measure of work** that can be **physically measured** at the **jobsite**, rather than using a **percentage**. For example, "**wire pulling**" can be easily measured in **linear feet** or "**concrete piers**" can be measured in **cubic yards**. However, some **precautions** must be used when **physical quantities** are used to represent **work in place of a percentage**. To illustrate, the **construction of concrete piers** involves **drilling, setting steel**, and **placement of concrete**. For a project that has **18 piers**, a progress report may show **all piers drilled, steel set in 9 piers**, and **concrete placed in 3 piers**. If **cubic yards** are the only measure of control, only **3 of the 18 piers** would be reported as complete, which would not include the **drilling and steel work** that is accomplished. For this situation, a **weight multiplier system** must be developed to account for each **task that is required to construct the piers**. As previously discussed, the **sum of all weights** must equal **1.0** to represent **100% of the work**.

在**最低级别**，可以使用在**工地上**可以**实际测量**的**工作计量单位**，而不是使用**百分比**。例如，"**电缆拉设**"可以用**线性英尺**来轻松测量，"**混凝土桩**"可以用**立方码**来测量。然而，当使用**物理量**来代表工作而不是百分比时，必须采取一些**预防措施**。例如，**混凝土桩的施工**包括**钻孔、钢筋安置**和**混凝土浇筑**。对于一个有 **18 根桩**的项目，进度报告可能显示**所有桩已钻孔，9 根桩已安置钢筋，3 根桩已浇筑混凝土**。如果只用**立方码**作为控制的唯一计量单位，那么仅会报告 **18 根桩中的 3 根已完成**，而未包括已完成的**钻孔和钢筋安置**。在这种情况下，必须开发一个**权重乘数系统**来涵盖构建桩所需的每个**任务**。如前所述，**所有权重之和**必须等于 **1.0**，以代表 **100%的工作**。

7. Calculate the **percent complete** for the project shown below using the **Weighted Units method** for measurement of work, similar to **Figure 13-13** in your book.

使用**加权单位法**计算下方项目的**完成百分比**，类似于书中的**图 13-13**。

Wt.	Subtask	U/M	Quant. total	Equivalent steel ton	Earned to date
0.02	Foundation bolts	Each	200	10.4	200
0.02	Shims	%	100	10.4	100
0.05	Shakeout	%	100	26.0	100
0.06	Columns	Each	84	31.2	80
0.10	Beams	Each	859	52.0	60
0.11	Cross-braces	Each	837	57.2	20
0.20	Girts & sag rods	Bay	38	104.0	10
0.09	Plumb & align	%	100	46.8	5
0.30	Connections	Each	2977	156.0	80
0.05	Punchlist	%	100	26.0	0
1.00	STEEL	TON		520.0	

FIGURE 13-13 Illustrative example of Weighted Units method for measurement of construction work (E4th Ch13 P501)

Measurement of Construction Work
施工工作的衡量

A **construction project** requires completion of numerous **tasks**, beginning with **initial clearing and site work** through the **final punchlist and cleanup**. Throughout the duration of a project, there must be a systematic **reporting of the progress of work** for each part of the project. The following paragraphs describe **six methods** for **measuring progress** during construction: **Units Completed**, **Incremental Milestone**, **Start/Finish**, **Supervisor Opinion**, **Cost Ratio**, and **Weighted Units**. The **system selected** depends on the **nature and complexity of the project** and the **desired level of control** by the **project manager**. Each of the six methods may be used on a given project. As stated before, it is important to define the **progress measurement method** for construction in the **PEP** and to use it **consistently during project construction**. Regular and consistent progress measurement helps to **identify trends** in **construction performance**. If **corrective actions** based on these trends occur in a **timely manner**, **adverse project outcomes** can be avoided.

一个**施工项目**需要完成许多**任务**，从**初步清理和场地工程**开始到**最终的验收清单和清理**。在整个项目的过程中，必须对项目各部分的**工作进度进行系统的报告**。以下段落描述了**六种方法**来衡量施工期间的**进度：完成单位法、增量里程碑法、开始/完成法、主管意见法、成本比率法**和**加权单位法**。所选择的**系统**取决于**项目的性质和复杂性**以及**项目经理所需的控制级别**。在一个项目中，可能会使用这六种方法中的每一种。如前所述，重要的是在**项目执行计划（PEP）中定义施工的进度测量方法**，并在项目施工期间**一致地使用**该方法。**定期且一致的进度测量**有助于**识别施工表现的趋势**。如果基于这些趋势的**纠正措施**能**及时采取**，则可以避免**不利的项目结果**。

The **U.S. Departments of Defense and Energy** have established what is known as the **Cost and Schedule Control Systems Criteria (C/SCSC)** for control of selected federal projects. While intended primarily for **high-value, cost-reimbursable research and development projects**, it may be applied to **selected construction projects**. Various methods for **measuring the status of a project** that have become common in the **construction industry** are described in **C/SCSC**.

美国国防部和能源部建立了一个称为**成本和进度控制系统标准（C/SCSC）的系统，用于控制特定的联邦项目。尽管该系统主要用于高价值、成本可报销的研究和开发项目**，但它也可以应用于**某些特定的施工项目**。**C/SCSC** 中描述了多种用于**衡量项目状态**的**方法**，这些方法已在**施工行业**中广泛应用。

The **Units Completed** method of measuring progress during construction is applicable to **tasks** that are **repetitive** and require a **uniform effort**. Generally, the **task** is the lowest level of control, so only one **unit of work** is necessary to define the work. To illustrate, the **percent complete** for **installation of wire** is determined as a **percentage**, by **dividing the number of feet installed** by the **total number of feet required**.

完成单位法适用于**重复性任务**，这些任务需要**统一的工作量**。一般来说，**任务**是最低的控制层级，因此只需一个**工作单位**来定义工作。例如，**电缆安装的完成百分比**是通过将**已安装的线缆长度**除以**所需安装的线缆总长度**来确定的。

The **Incremental Milestone** method is applicable to tasks that include **subtasks** handled in **sequence**. For example, the **installation of a major vessel** in an **industrial facility** may include **sequential tasks** as shown

below. **Completion** of any subtask is considered a **milestone**, representing a **certain percentage** of the total installation. The **percentage** may be based on the **estimated work-hours** needed to accomplish the work.
增量里程碑法适用于包含按**顺序进行的子任务**的任务。例如，在**工业设施中安装主要设备**可能包含以下**顺序任务**。完成任何子任务被视为一个**里程碑**，代表**总安装量的某个百分比**。该百分比可以基于完成工作的**估计工时**来确定。

Received and inspected	15%
Setting complete	35%
Alignment complete	50%
Internals installed	75%
Testing complete	90%
Accepted by owner	100%

The **Start/Finish** method is applicable to tasks that do not have **well-defined intermediate milestones** or for which the **time required** is difficult to estimate. For example, **alignment of a piece of equipment** may take a **few hours to a few days**, depending on the situation. The workers may know when the work will **start** and when it is **finished** but never know the **percentage completion in between**. For this method, an **arbitrary percent complete** is assigned at the **start of the task**, and **100% complete** is assigned when it is finished. A starting percentage of **20% to 30%** might be assigned for tasks that require a **long duration**, while **0%** may be assigned for tasks with a **short duration**.

开始/完成法适用于那些**没有明确的中间里程碑**或**所需时间难以估算**的任务。例如，**设备对准**的时间可能从**几小时到几天**不等，视情况而定。工人可能知道工作**何时开始**和**何时结束**，但无法了解中间的**完成百分比**。对于此方法，在任务开始时**分配一个任意的完成百分比**，并在任务结束时分配 **100%的完成度**。对于持续时间较长的任务，可能分配 **20%到 30%的初始百分比，而对于持续时间较短的任务，可能分配 0%**。

The **Supervisor Opinion** method is a **subjective approach** used for **minor tasks**, such as **construction support facilities**, where a more **discrete method** cannot be applied.

主管意见法是一种**主观方法**，可用于**次要任务**，如**施工支持设施**，此类任务难以使用更精细的方法进行衡量。

The **Cost Ratio** method is applicable for **administrative tasks**, such as **project management, quality assurance, contract administration**, or **project control**. These tasks involve a **long period of time** or are **continuous** throughout the duration of the project. Generally, these tasks are **estimated and budgeted** as **lump-sum dollars** and **work-hours**, rather than **measurable quantities of production work**. For this method, the **percent complete** can be calculated by a specific equation.

成本比率法适用于**管理任务**，例如**项目管理、质量保证、合同管理**或**项目控制**。这些任务的持续时间较长或贯穿项目始终。通常，这些任务被**估算和预算为总额（金额和工时）**，而不是**生产工作的可量化数量**。对于此方法，可以通过特定方程计算**完成百分比**。

$$\text{Percent complete} = \frac{\text{Actual cost or work-hours to date}}{\text{Forecast at completion}}$$

The **Weighted Units** method applies to tasks that involve **major efforts** over a **long duration of time**. Generally, the work requires several **overlapping subtasks**, each with a different **unit of measurement**. This method is illustrated by the **structural steel example** shown in **Figure 13-13**. A common **unit of measure** for **steel** is **tons**. A **weight** is assigned to each subtask to represent the **estimated level of effort**. **Work-hours** is often a good measure of the required **level of effort**. As **quantities of work** are completed for each subtask, they are **converted into equivalent tons** and **percent complete** is calculated as shown in **Figure 13-13**.

加权单位法适用于那些涉及**大量工作**且**持续时间较长**的任务。通常，这类工作需要**多个重叠的子任务**，每个子任务有不同的**工作单位**。**图 13-13** 中的**结构钢**示例说明了该方法。钢材的常用**度量单位**是**吨**。为每个子任务分配一个**权重**以代表**估计的工作量**。**工时**通常是衡量所需**工作量**的良好指标。随着每个子任务**工作量的完成**，这些数量会被**转换为等效吨数**，并按**图 13-13** 所示计算**完成百分比**。

Wt.	Subtask	U/M	Quantity Total	Equivalent Steel Ton	Quantity to-date	Earned Tons*
0.02	Run foundation bolts	each	200	10.4	200	10.4
0.02	Shim	%	100	10.4	100	10.4
0.05	Shakeout	%	100	26.0	100	26.0
0.06	Columns	each	84	31.2	74	27.5
0.10	Beams	each	859	52.0	0	0.0
0.11	Cross-braces	each	837	57.2	0	0.0
0.20	Girts & sag rods	bay	38	104.0	0	0.0
0.09	Plumb & align	%	100	46.8	5	2.3
0.30	Connections	each	2977	156.0	74	3.9
0.05	Punch list	%	100	26.0	0	0.0
1.00	STEEL	TON		520.0		80.5

$$^{*}\text{Earned Tons to Date} = \frac{(\text{Quantity to date}) \text{ X } (\text{Relative Weight})}{(\text{Total Quantity})}$$

$$\text{Percent complete} = \frac{80.5 \text{ tons}}{520 \text{ tons}} = 15.5\%$$

FIGURE 13-13 Illustrative example of Weighted Units method for measurement of construction work. (Source: **Construction Industry Institute**, Publication No. 6-5.)
图 13-13 使用加权单位法衡量施工工作的示例。（来源：**建筑业协会**，出版物第 6-5 号。）

As discussed, there are many **ways of measuring progress** of work for each task in a project. After determining the **progress of different work tasks**, the next step is to develop a **method that combines the tasks** to determine the **overall percent complete** for the project. The **earned-value system** can be used to define the **overall percent complete** for the **entire project**. **Earned value** can be linked to the **project budget**, expressed in **work-hours** or **dollars**. For a **single account**, the **earned value** can be calculated by the following equation:

如前所述，有多种**方法**可以**衡量项目中每项任务的工作进度**。在确定不同任务的进度后，下一步是开发一种**方法**来将这些任务**结合起来**，以确定项目的**整体完成百分比**。**挣值系统**可以用于定义**整个项目的整体完成百分比**。**挣值**可以与项目**预算**关联，预算以**工时**或**金额**表示。对于单个账户，可以通过以下公式计算**挣值**：

Earned value = (**Percent complete**) × (**Budget for that account**)

挣值 = （**完成百分比**） × （**该账户的预算**）

A **budgeted amount** is “earned” as a task is completed, up to the **total amount in that account**. For example, an account may be **budgeted at $10,000 and 60 work-hours**. If the account is reported as **25% complete**, as measured by one of the previously described methods, then the **earned value** is **$2,500 and 15 work-hours**. Thus, progress in all accounts can be reduced to **earned work-hours and dollars**, providing a **method for summarizing multiple accounts** and calculating **overall progress** for the **total project**. The equation for determining the **overall project percent complete** is given below.

当一个任务完成时，**预算金额**被“挣得”，直到达到该账户的**总金额**。例如，一个账户可能预算了 **10,000 美元和 60 工时**。如果该账户报告为**完成 25%**（按照前述方法之一衡量），那么**挣值**为 **2,500 美元和 15 工时**。因此，所有账户的进展可以简化为**挣得的工时和金额**，从而为**汇总多个账户和计算整个项目的整体进展**提供方法。以下是确定**整个项目完成百分比**的方程式。

$$\text{Percent complete} = \frac{(\text{Earned work-hours or dollars all accounts})}{(\text{Budgeted work-hours or dollars all accounts})}$$

The concepts discussed above provide a **system** for determining the **percent complete** of a **single work task** or **combinations of tasks**. This provides the **basis for analyzing results** to determine **how well work is proceeding** compared to what was **planned**. The **earned-value system** provides a **method for evaluating performance** of the project.

以上概念提供了一个**系统**来确定**单一工作任务**或**多个任务组合**的**完成百分比**。这为**分析结果**提供了基础，以确定工作与**计划**相比**进行得如何**。**挣值系统**提供了一种评估项目**表现的方法**。

The previous paragraphs have only discussed **budgeted and earned work-hours and dollars**. **Actual work-hours and dollars** must also be included to evaluate the **performance of the project**. The following definitions describe the procedure for **evaluating cost and schedule performance**:

前面仅讨论了**预算和挣得的工时和金额**。还必须包含**实际工时和金额**，以便评估**项目的表现**。以下定义描述了**评估成本和进度表现**的过程：

- **Budgeted work-hours or dollars to date** represent **what is planned to do**. The **C/SCSC** defines this as **Budgeted Cost for Work Scheduled (BCWS)**.
- **Earned work-hours or dollars to date** represent **what was done**. The **C/SCSC** defines this as **Budgeted Cost for Work Performed (BCWP)**.
- **Actual work-hours or dollars to date** represent **what has been paid**. The **C/SCSC** defines this as **Actual Cost of Work Performed (ACWP)**.

- **截止当前日期的预算工时或金额**代表**计划要做的内容**。**C/SCSC** 将其定义为**计划工作预算成本**（**BCWS**）。
- **截止当前日期的挣得工时或金额**代表**实际完成的工作**。**C/SCSC** 将其定义为**实际完成工作的预算成本**（**BCWP**）。
- **截止当前日期的实际工时或金额**代表**已支付的成本**。**C/SCSC** 将其定义为**实际完成工作的实际成本**（**ACWP**）。

Performance against schedule is a **comparison of what was planned against what was done**, i.e., a **comparison of budgeted and earned work-hours**. If the **budgeted work-hours** are less than the **earned work-hours**, it means **more was done than planned**, and the project is **ahead of schedule**. The opposite would indicate the project is **behind schedule**.

进度表现是**计划内容与实际完成内容**的对比，即**预算工时和挣得工时的对比**。如果**预算工时**小于**挣得工时**，说明完成量超过计划，项目**进度超前**。反之则表示**进度落后**。

Performance against budget is measured by comparing **what was done** to **what was paid**. This compares **earned work-hours** to **actual work-hours** or **cost**. If **more was paid than is done**, then the project would have a **cost overrun**. The following **variance** and **index equations** can be used to calculate these values:

预算表现通过**完成的工作量**与**已支付的费用**进行对比来衡量。这将**挣得工时**与**实际工时或成本**进行对比。如果**支付的费用**超过**完成的工作量**，则项目存在**成本超支**。以下**差异**和**指数公式**可以用于计算这些值：

$$SV = BCWP - BCWS$$

$$SPI = \frac{\text{Earned work-hours or dollars to date}}{\text{Budgeted work-hours or dollars to date}}$$

$$SPI = \frac{BCWP}{BCWS}$$

Cost variance (CV) = **(Earned work-hours or dollars)** − **(Actual work-hours or dollars)**

成本差异（CV） = **（挣得工时或金额）** − **（实际工时或金额）**

$$CV = BCWP - ACWP$$

$$CPI = \frac{\text{Earned work-hours or dollars to date}}{\text{Actual work-hours or dollars to date}}$$

$$CPI = \frac{BCWP}{ACWP}$$

A **positive variance** and an **index of 1.0 or greater** indicate **favorable performance**.

正差异和**指数为 1.0 或更大**表示**表现良好**。

8. The **EPC project** shown in **Appendix A** has an **original planned schedule** of **17 months** with a **budget**, excluding contingency, of **$45,797,250**. Below are **update reports** for the **first 4 months** of the project.

附录 A 中显示的 **EPC 项目**的**原计划进度**为 **17 个月**，预算（不含应急费用）为 **45,797,250 美元**。以下是该项目**前 4 个月**的**更新报告**。

Status report for month	Planned (BCWS)	Earned (BCWP)	Actual (ACWP)
1	$58,395	$48,000	$47,000
2	$685,080	$720,000	$750,000
3	$2,773,560	$2,695,000	$3,125,000
4	$4,734,225	$4,780,000	$5,235,000

Perform an **earned value analysis** of the project, similar to **Example 13-2** in your book. For each of the **4 months**, calculate the **variances (CV and SV)**, the **performance indices (CPI and SPI)**, the **estimate to complete (ETC)**, and the **estimate at completion (EAC)**. Prepare a **graph** similar to **Figure 13-15** to show trends in the **project performance**. Use **Figure 13-15** to guide your interpretation of the **CPI and SPI** for each reporting period.

对该项目执行**挣值分析**，类似于书中的**示例 13-2**。对于每个 **4 个月**，计算**差异（CV 和 SV）、绩效指数（CPI 和 SPI）、完成估算（ETC）和最终估算（EAC）**。请绘制一个**类似于图 13-15** 的图表，以显示**项目绩效趋势**。使用**图 13-15** 来指导对每个报告期的 **CPI 和 SPI** 的解释。

$$BAC = \text{Original project estimate} \qquad \text{Budget at completion}$$

$$ETC = \frac{BAC - BCWP}{CPI} \qquad \text{Estimate to complete}$$

$$EAC = ACWP + ETC \qquad \text{Estimate at completion}$$

Ratios are used in the earned-value system to predict the cost to complete a project. The CPI is used to predict the magnitude of a possible cost overrun or underrun. It adjusts the budget based on past performance. The SPI is used to predict the magnitude of a possible time advance or delay. It adjusts the schedule based on past performance. The following examples illustrate the calculations in an earned-value analysis. The examples provide a progressive analysis of the project with an interpretation of the results based on the earned-value system for assessing project performance.

Example 13-2 The EPC project shown in Appendix A has an original planned schedule of 17 months with a budget of $45,797,250, excluding contingency (see Figure A-3). Figure A-10 shows the monthly distribution of costs for all activities of the project. The costs are shown on an early start, late start, and target schedule basis. The target schedule is used for planned BCWS cost in this earned value analysis. Below are update reports for the first 8 months of the project.

Status report for month	Planned (BCWS)	Earned (BCWP)	Actual (ACWP)
1	$58,935	$58,050	$59,520
2	$685,080	$643,980	$670,815
3	$2,773,560	$2,440,725	$2,871,435
4	$4,734,225	$4,639,545	$5,272,215
5	$7,190,145	$7,621,545	$7,857,270
6	$8,187,945	$8,351,700	$8,030,475
7	$10,008,150	$10,708,725	$9,915,480
8	$12,701,100	$12,193,050	$10,790,310

Perform an earned-value analysis of the project. For each report period, calculate the variances (CV and SV), the performance indices (CPI and SPI), the estimate to complete (ETC), and the estimate at completion (EAC). Prepare a graph similar to Figure 13-15 to show trends in the project performance. Figures 13-16 and 13-17 provide a guide to interpret the CPI and SPI for each reporting period. The first month's analysis is shown below.

Analysis of Month #1:
The status report after Month #1 into the project includes the following information:

$$\begin{aligned} \text{Planned, BCWS} &= \$58,935 \\ \text{Earned, BCWP} &= \$58,050 \\ \text{Actual, ACWP} &= \$59,520 \\ \text{Budget at Completion, BAC} &= \$45,797,250 \end{aligned}$$

Cost and Schedule Deviations:

$$\begin{aligned} \text{Cost Variance} &= \text{BCWP} - \text{ACWP} \\ &= \$58,050 - \$59,520 \\ &= -\$1,470 \end{aligned}$$

A negative value of CV represents a cost overrun. Based on the status report, the actual cost is greater than earned by $1,470.

$$\begin{aligned} \text{Schedule Variance} &= \text{BCWP} - \text{BCWS} \\ &= \$58,050 - \$58,935 \\ &= -\$885 \end{aligned}$$

A negative value of SV represents a schedule slippage. The project is behind the planned schedule.
Cost and Schedule Performance:

$$\begin{aligned} \text{Cost Performance Index, CPI} &= \text{BCWP}/\text{ACWP} \\ &= \$58,050/\$59,520 \\ &= 0.98 \end{aligned}$$

The CPI is less than 1, which indicates a poor cost performance. The earned value is less than the actual costs.

$$\begin{aligned} \text{Schedule Performance Index, SPI} &= \text{BCWP}/\text{BCWS} \\ &= \$58,050/\$58,935 \\ &= 0.98 \end{aligned}$$

The SPI is less than 1, which indicates the schedule performance is worse than planned. The project is behind schedule.

Forecasting Cost at Completion

$$\begin{aligned} \text{Estimate to Complete, ETC} &= (\text{BAC} - \text{BCWP})/\text{CPI} \\ &= (\$45,797,250 - \$58,050)/0.98 \\ &= \$46,672,653 \end{aligned}$$

Based on the analysis of the status report, the remaining cost to complete the project is \$46,672,653.

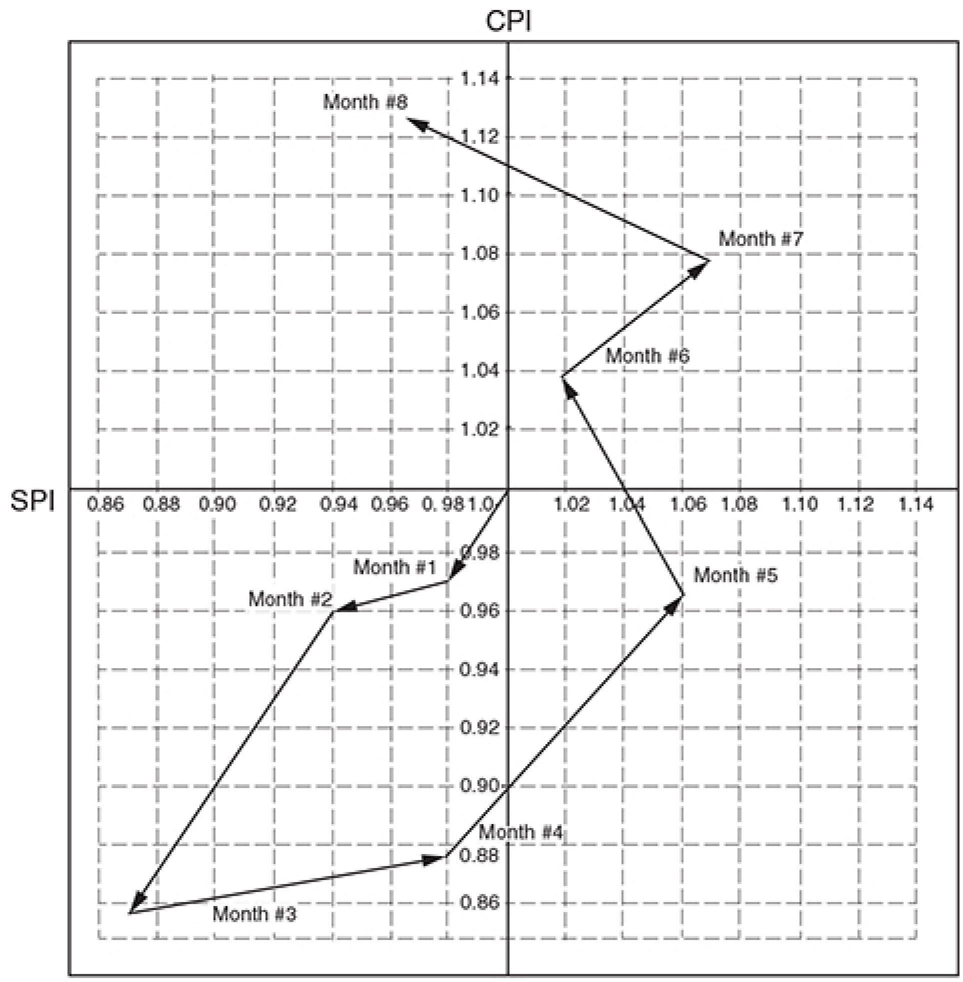
$$\begin{aligned}\text{Estimate at Completion, EAC} &= \text{ACWP} + \text{ETC}\\ &= \$59{,}520 + \$46{,}672{,}653\\ &= \$46{,}732{,}173\end{aligned}$$

Based on analysis of the status report, the estimated cost of the project at completion is \$46,732,173, which is \$934,653 over the original budget of \$45,797,520.

Following is a summary of the earned-value analysis and a project performance chart comparing CPI to SPI for the 8 months. SPI is plotted on the horizontal axis; CPI is plotted on the vertical axis.

Earned Value Analysis by Month

Month	CV = BCWP – ACWP	SV = BCWP – BCWS	CPI = BCWP / ACWP	SPI = BCWP / BCWS	ETC = (BAC – BCWP) / CPI	EAC = ACWP + ETC
Month #1	–\$1,470	–\$885	0.98	0.98	\$46,672,653	\$46,732,173
Month #2	–\$26,835	–\$41,100	0.96	0.94	\$47,034,656	\$47,705,471
Month #3	–\$430,710	–\$332,835	0.85	0.88	\$51,007,676	\$53,879,111
Month #4	–\$632,670	–\$94,680	0.88	0.98	\$46,770,119	\$52,042,334
Month #5	–\$235,725	\$431,400	0.97	1.06	\$39,356,397	\$47,213,667
Month #6	\$321,225	\$163,755	1.04	1.02	\$36,005,337	\$44,035,812
Month #7	\$793,245	\$700,575	1.08	1.07	\$32,489,375	\$42,404,855
Month #8	\$1,402,740	–\$508,050	1.13	0.96	\$29,738,230	\$40,528,540



Project Performance—CPI and SPI

Example 13-3 Provide an earned-value analysis to evaluate the progress of the sewer and water lines project shown in Figure 10-6. The original budget is $147,500 and the project is scheduled to be completed in 94 working days. A status report after 10 working days into the project includes the following information:

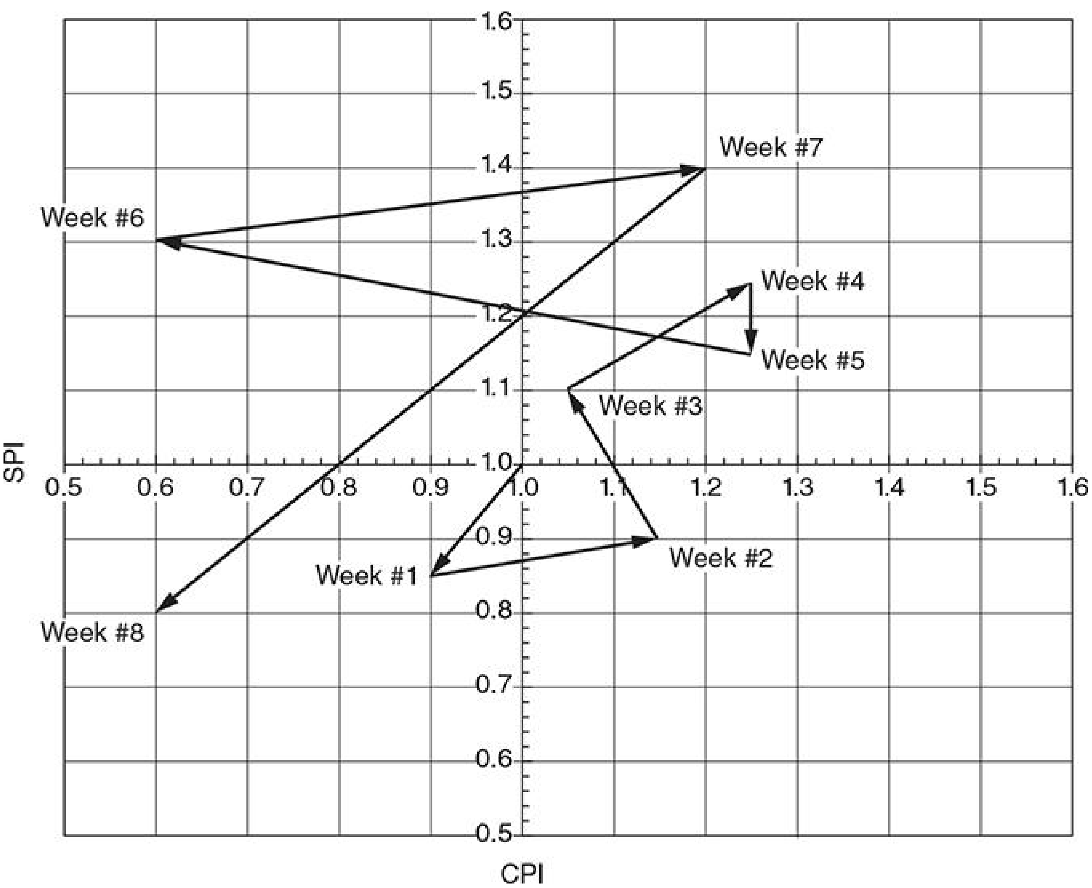
FIGURE 13-15 Monitoring project performance—the CPI and SPI (E4th Ch13 P518)

Monitoring Project Performance
监控项目绩效

The **CPI** and **SPI** provide a **quantitative measurement** of the **progress of a project**. Values of **CPI** or **SPI** that are **greater than 1.0** indicate **good project performance**, whereas values of **CPI** or **SPI** that are **less than 1.0** indicate **poor project performance**. Values of these indices can be **plotted on a graph**, as shown in **Figure 13-15**, to evaluate **project performance** during routine **reporting periods**. The greatest value of measuring **CPI** and **SPI** regularly during a project is **identifying trends early**. **Unexplained trends** or **sudden changes** should alert the **project manager** to **investigate** and take **corrective action** if warranted.

CPI 和 **SPI** 为**项目进度**提供了**定量衡量**。**CPI** 或 **SPI** 的值**大于 1.0** 表明**项目表现良好**，而 **CPI** 或 **SPI** 的值**小于 1.0** 则表明**项目表现不佳**。这些指标的数值可以绘制在**图表**上，如**图 13-15** 所示，以评估项目在例行**报告期**中的**表现**。在项目过程中定期衡量 **CPI** 和 **SPI** 的最大价值在于能够**及早识别趋势**。如果出现**无法解释的趋势**或**突然变化**，应提醒**项目经理**进行**调查**并在必要时采取**纠正措施**。

FIGURE 13-15 Monitoring project performance—the CPI and SPI.
图 13-15 监控项目绩效—CPI 和 SPI



As shown in **Figure 13-15**, a project begins with a **CPI and SPI equal to 1.0**. Projects are **dynamic in nature** due to **events that can impact the cost and the schedule**. Therefore, it is expected that these **performance indices** will **deviate from this initial starting point of 1.0** as the project progresses **week to week**. The **project manager** should not **overreact** to slight changes in the **CPI and SPI**. For example, the **first 2 weeks** of the project show **small changes** in the **performance indices**, which might be expected during the **early progress** of the project.

如**图 13-15** 所示，一个项目的起始 **CPI 和 SPI** 值为 **1.0**。项目由于**会影响成本和进度的事件**，其性质是**动态的**。因此，随着项目的**逐周推进**，预计这些**绩效指标**会**偏离 1.0 的初始值**。**项目经理**不应对 **CPI 和 SPI 的轻微变化反应过度**。例如，项目的**前两周**显示了**绩效指标的轻微变化**，这在项目的**初期进展**中是可以预料的。

Although **minor deviations** are expected, **major deviations** from one **report period to the next** should alert the **project manager** to **investigate** the reason for the **significant change in project performance**. For example, the report between **weeks 7 and 8** shows a **dramatic change toward poor project performance**. This type of report merits the **attention of the project manager** to solicit **team members** to identify **what has happened** in the project and the **reason for this dramatic change** in project performance.

尽管**轻微偏差**是正常的，但**从一个报告期到下一个报告期**的**重大偏差**应引起**项目经理的警觉**，需要**调查**导致项目表现**显著变化**的原因。例如，**第 7 周至第 8 周**的报告显示出**项目表现急剧恶化**。这种报告值得**项目经理关注**，并与**团队成员**一起找出项目中发生的**原因**及其对项目表现的影响。

Concern should also be raised if the **trend of the CPI and SPI** is progressing toward **poor performance**. For example, from **week 2 through week 4**, the project performance trend is **toward good**. However, **significant changes** in the project begin to appear during **weeks 6, 7, and 8**.

如果 **CPI 和 SPI 的趋势**走向**不佳表现**，也应引起关注。例如，从**第 2 周到第 4 周**，项目表现的趋势是**向好**的。然而，**显著变化**在**第 6 周、第 7 周和第 8 周**开始显现。

Thus, the **project manager** must be **vigilant** regarding changes in the project. The **CPI and SPI** provide some of the information needed by the **project manager** to **monitor the progress of work** to ensure the project is moving in the **right direction toward successful completion**.

因此，**项目经理**必须**密切关注**项目中的变化。**CPI 和 SPI** 提供了项目经理所需的部分信息，以**监控工作的进展**，确保项目朝着**成功完成的正确方向推进**。

2nd Ch9Q8

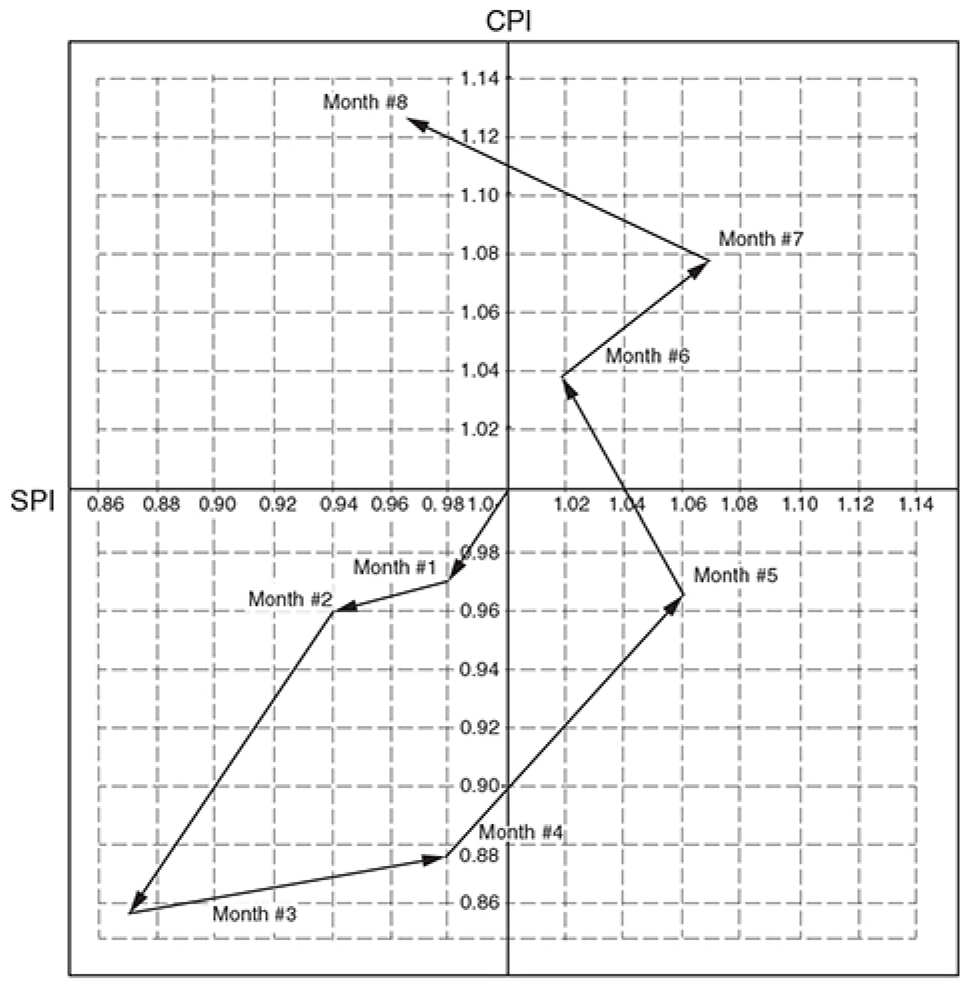
The EPC project shown in **Appendix A** has an **original planned schedule** of **17 months** with a **budget**, excluding contingency, of **$3,053,150**. Below are **update reports** for the **first 8 months** of the project.

附录 A 中显示的 **EPC 项目**的**原计划进度**为 **17 个月**，预算（不含应急费用）为 **3,053,150 美元**。以下是该项目**前 8 个月**的**更新报告**。

Status report for month	Planned (BCWS)	Earned (BCWP)	Actual (ACWP)
1	$3,929	$3,870	$3,968
2	$45,672	$42,932	$44,721
3	$184,904	$162,715	$191,429
4	$315,615	$309,303	$351,481
5	$479,343	$508,103	$523,818
6	$545,863	$556,780	$535,365
7	$667,210	$713,915	$661,032
8	$846,740	$812,870	$719,354

Perform an earned-value analysis of the project, similar to **Examples 9-1 through 9-4**. For each **report period**, calculate the **variances (CV and SV)**, the **performance indices (CPI and SPI)**, the **estimate to complete (ETC)**, and the **estimate at completion (EAC)**. Prepare a **graph** similar to **Figure 9-14** to show **trends in the project performance**. Use **Figure 9-15** to guide your **interpretation of the CPI and SPI** for each reporting period.

对该项目**执行挣值分析**，类似于书中的**示例 9-1 至 9-4**。对于每个**报告期**，计算**差异（CV 和 SV）**、**绩效指数（CPI 和 SPI）**、**完成估算（ETC）和最终估算（EAC）**。请绘制一个**类似于图 9-14 的图表**，以显示**项目绩效趋势**。使用**图 9-15** 来指导**对每个报告期的 CPI 和 SPI 的解释**。



Project Performance—CPI and SPI

Example 13-3 Provide an earned-value analysis to evaluate the progress of the sewer and water lines project shown in Figure 10-6. The original budget is $147,500 and the project is scheduled to be completed in 94 working days. A status report after 10 working days into the project includes the following information:

Water & Sewer Line Project Status after 10 working days			
BAC: $147,500		**Schedule:** 94 working days	
Current Period: 10 working days			
Activity ID	**% Complete**	**Schedule**	**Actual Cost**
10	100%	Complete as scheduled	$1,500
20	100%	Complete as scheduled	$2,200
30	100%	Complete as scheduled	$4,000
BCWP	**ACWP**	**BCWS**	
$7,600	$7,700	$7,600	

The BCWS and BCWP shown above are the same values as shown on the 10th working day in Figure 10-12 because Activities 10, 20, and 30 were all completed according to the original planned schedule.

The earned-value analysis is summarized below.

Earned Value Analysis for Sewer & Water Line Project after 10 working days		
CV = BCWP – ACWP	–$100	Negative CV represents a cost overrun. Based on the status report, the actual cost is greater than the earned cost by $100.
SV = BCWP – BCWS	0	SV is 0 which means the project is progressing as planned. The project is not ahead or behind schedule.
CPI = BCWP/ACWP	0.987	CPI is less than 1.0, indicating a poor cost performance. The earned value is less than the actual costs.
SPI = BCWP/BCWS	1.000	SPI is equal to 1.0, indicating the schedule performance is progressing according to plan.
ETC = (BAC – BCWP)/CPI	$141,743	This is the remaining cost, based on performance to date, to complete the project.
EAC = ACWP + ETC	$149,443	Based on the analysis of the status report, the project cost at completion will exceed the project's budgeted cost.
(Over)/Under Budget	–$1,943	

Example 13-4 Provide an earned-value analysis to evaluate the progress of the sewer and water lines project shown in Figure 10-6. The original budget is $147,500 and the project is scheduled to be completed in 94 working days. A status report after 20 working days into the project includes the following information:

Water & Sewer Line Project Status after 20 working days			
BAC: $147,500		**Schedule:** 94 working days	
Current Period: 20 working days			
Activity ID	**% Complete**	**Schedule**	**Actual Cost**
40	100%	2 days delayed	$8,000
50	60%	2 days delayed	$3,200
60	50%	Out of sequence start	$6,300
	BCWP	**ACWP**	**BCWS**
Current status:	$22,720	$25,200	$22,400
Previous status:	$7,600	$7,700	$7,600

The BCWS shown above is different than the value shown on the 20th working day in Figure 10-12 because Activities 40, 50, and 60 have been completed at different times than the original planned schedule. The BCWS of $22,400 shown above is based on an updated schedule for the project. Also, the BCWP is based on the updated schedule.

The earned-value analysis for the project is summarized below.

Earned Value Analysis for Sewer & Water Line Project after 20 working days		
CV = BCWP – ACWP	–$2,480	Negative CV represents a cost overrun. Based on the status report, the actual cost is greater than the earned cost by $2,480.
SV = BCWP – BCWS	$320	SV is $320 which means the earned value is greater than planned. The project has greater than planned effort and may be ahead of schedule.
CPI = BCWP/ACWP	0.902	CPI is less than 1.0, indicating a poor cost performance. The earned value is less than the actual costs.
SPI = BCWP/BCWS	1.014	SPI is greater than 1.0, indicating the schedule performance is progressing better than plan.
ETC = (BAC – BCWP)/CPI	$138,337	This is the remaining cost, based on performance to date, to complete the project.
EAC = ACWP + ETC	$163,537	Based on the analysis of the status report, the project cost at completion will exceed the project's budgeted cost and is significantly worse than the status of Example 13-3.
(Over)/Under Budget	–$16,037	

Example 13-5 Provide an earned-value analysis to evaluate the progress of the sewer and water lines project shown in Figure 10-6. The original budget is $147,500 and the project is scheduled to be completed in 94 working days. A status report after 25 working days into the project includes the following information:

Water & Sewer Line Project Status after 25 working days

BAC: $147,500 **Schedule:** 94 working days

Current Period: 25 working days

Activity ID	% Complete	Schedule	Actual Cost
50	100%	1 more day delay	$6,200
60	100%	No additional delays	$12,600
70	100%	Complete, no delays	$2,100
	BCWP	**ACWP**	**BCWS**
Current status:	$33,000	$36,600*	$31,740
Previous status:	$22,720	$25,200	$22,400

*Includes final actual costs of Activities 10–70

The earned-value analysis for this status report is summarized below.

Earned Value Analysis for Sewer & Water Line Project after 25 working days		
CV = BCWP – ACWP	–$3,600	Negative CV represents a cost overrun. Based on the status report, the actual cost is greater than the earned cost by $3,600.
SV = BCWP – BCWS	$1,260	SV is $1,260 which means the earned value is greater than planned. The project has greater than planned effort and may be ahead of schedule.
CPI = BCWP/ACWP	0.902	CPI is less than 1.0, indicating a poor cost performance. The earned value is less than the actual costs.
SPI = BCWP/BCWS	1.040	SPI is greater than 1.0, indicating the schedule performance is progressing better than plan.
ETC = (BAC – BCWP)/CPI	$126,940	This is the remaining cost, based on performance to date, to complete the project.
EAC = ACWP + ETC	$163,540	Based on the analysis of the status report, the project cost at completion will exceed the project's budgeted cost and is about the same as Example 13-4.
(Over)/Under Budget	–$16,040	

Example 13-6 Provide an earned-value analysis to evaluate the progress of the sewer and water lines project shown in Figure 10-6. The original budget is $147,500 and the project is scheduled to be completed in 94 working days. A status report after 70 working days into the project includes the following information:

Water & Sewer Line Project Status after 70 working days			
BAC: $147,500		**Schedule:** 94 working days	
Current Period: 70 working days			
Activity ID	**% Complete**	**Schedule**	**Actual Cost**
80	100%	Complete, no delays	$14,000
90	100%	Complete, no delays	$800
100	100%	Complete, no delays	$1,400
110	100%	Complete, no delays	$5,000
130	100%	Complete, no delays	$4,500
140	20%	No delays	$680
160	30%	No delays	$2,100
	BCWP	**ACWP**	**BCWS**
Current status:	$56,830	$65,080	$102,800
Previous status:	$33,000	$36,600	$31,740

The earned-value reporting for this project status is provided below.

Earned Value Analysis for Sewer & Water Line Project after 70 working days		
CV = BCWP – ACWP	–$8,250	Negative CV represents a cost overrun. Based on the status report, the actual cost is greater than the earned cost by $3,600.
SV = BCWP – BCWS	–$45,970	SV is a negative value of $45,970 which means the earned value is less than planned. The project has less than planned effort and may be behind schedule.
CPI = BCWP/ACWP	0.873	CPI is less than 1.0, indicating a poor cost performance. The earned value is less than the actual costs.
SPI = BCWP/BCWS	0.553	SPI is greater than 1.0, indicating the schedule performance is progressing better than plan.
ETC = (BAC – BCWP)/CPI	$103,860	This is the remaining cost, based on performance to date, to complete the project.
EAC = ACWP + ETC	$168,940	Based on the analysis of the status report, the project cost at completion will exceed the project's budgeted cost by $21,440 which is a significant cost overrun.
(Over)/Under Budget	–$21,440	

Figure 13-14 shows the variation by reporting period of CPI and SPI for Examples 13-3 through 13-6. CPI and SPI can also be compared similar to Example 2 in order to show project performance.

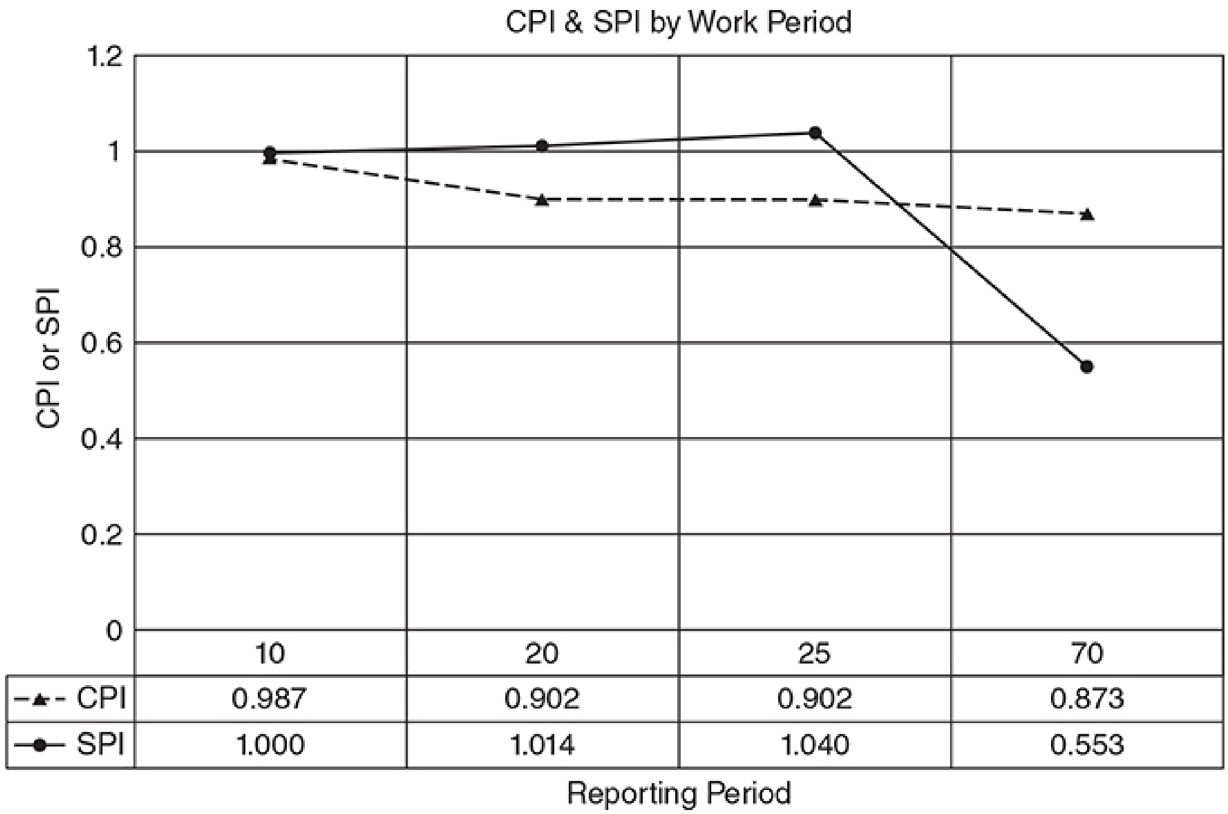


FIGURE 13-14 CPI and SPI variation by reporting period for example problems 13-3 through 13-6.

Monitoring Project Performance

The CPI and SPI provide a quantity measurement of the progress of a project. Values of CPI or SPI that are greater than 1.0 indicate good project performance, whereas values of CPI or SPI that are less than 1.0 indicate poor project performance. Values of these indices can be plotted on a graph as shown in Figure 13-15 to evaluate project performance during routine reporting periods. The greatest value of measuring CPI and SPI regularly during a project is identifying trends early. Unexplained trends or sudden changes should alert the project manager to investigate and take corrective action if warranted.

FIGURE 13-16 Interpretations of Cost Performance Index (CPI) and Schedule Performance Index (SPI) (E4th Ch13 P520)

Same as 2nd Ch9 Figure 9-15

Interpretation of Performance Indices
绩效指标的解读

The **graph** shown in **Figure 13-15** is a valuable tool for the **project manager** to monitor **project performance** based on **progress reports**. It provides a **measure** of how well the **planned cost** compares to **actual expenditures** and **work accomplished**. As **deviations** occur between the values, several **interpretations** can be made.

图 13-15 所示的**图表**是**项目经理**根据**进度报告**监控**项目绩效**的宝贵工具。它提供了一种**衡量**手段，用来比较**计划成本**与**实际支出**及**完成的工作量**的表现。当这些数值出现**偏差**时，可以进行多种**解读**。

Values of **SPI** greater than **1.0** indicate the project is **ahead of schedule**. The project may be progressing **faster than expected** if the original **production rates were predicted too low** or the **actual working conditions** are better than originally anticipated. There may be **more staffing** on the project than anticipated, which would also show the work as progressing **ahead of schedule**. Values of **SPI** less than **1.0** indicate the project is **behind schedule**. The project schedule may **slip** due to **weather delays, understaffing**, or **disorganized work**.

SPI 值**大于 1.0** 表明项目**进度超前**。如果最初的**生产率预测偏低**或**实际工作条件**优于预期，项目可能进展得**比预期更快**。如果项目**人员配备**超出预期，也会表现出**进度超前**的迹象。**SPI** 值**小于 1.0** 表明项目**进度落后**。项目进度可能因**天气延误、人手不足**或**工作组织不当**而**延迟**。

Values of **CPI** greater than **1.0** indicate **good project cost performance**. Good cost performance may be reported if **actual productivity** is better than planned or if the **measured percent of completed work** is too high. Values of **CPI** less than **1.0** indicate **poor cost performance**, which may result from **poorer-than-planned productivity** or an **underestimate of the measured percent of completed work**.

CPI 值**大于 1.0** 表明项目的**成本表现良好**。如果**实际生产率**优于计划或**已完成工作的百分比**高于实际，成本表现会更好。**CPI** 值**小于 1.0** 表明**成本表现不佳**，可能是由于**生产率低于预期**或**已完成工作的测量百分比估计不足**所导致的。

The **project manager** should seek **information from team members** to provide a better **interpretation** of the meaning of **project performance indices**. **Figures 13-16** and **13-17** provide additional **interpretations of CPI and SPI data**.

项目经理应向**团队成员**寻求更多信息，以更好地**解读项目绩效指标**的含义。**图 13-16** 和**图 13-17** 提供了更多关于 **CPI 和 SPI 数据**的**解读**。

FIGURE 13-16 Interpretations of Cost Performance Index (CPI) and Schedule Performance Index (SPI).

图 13-16 成本绩效指数（CPI）和进度绩效指数（SPI）的解读。

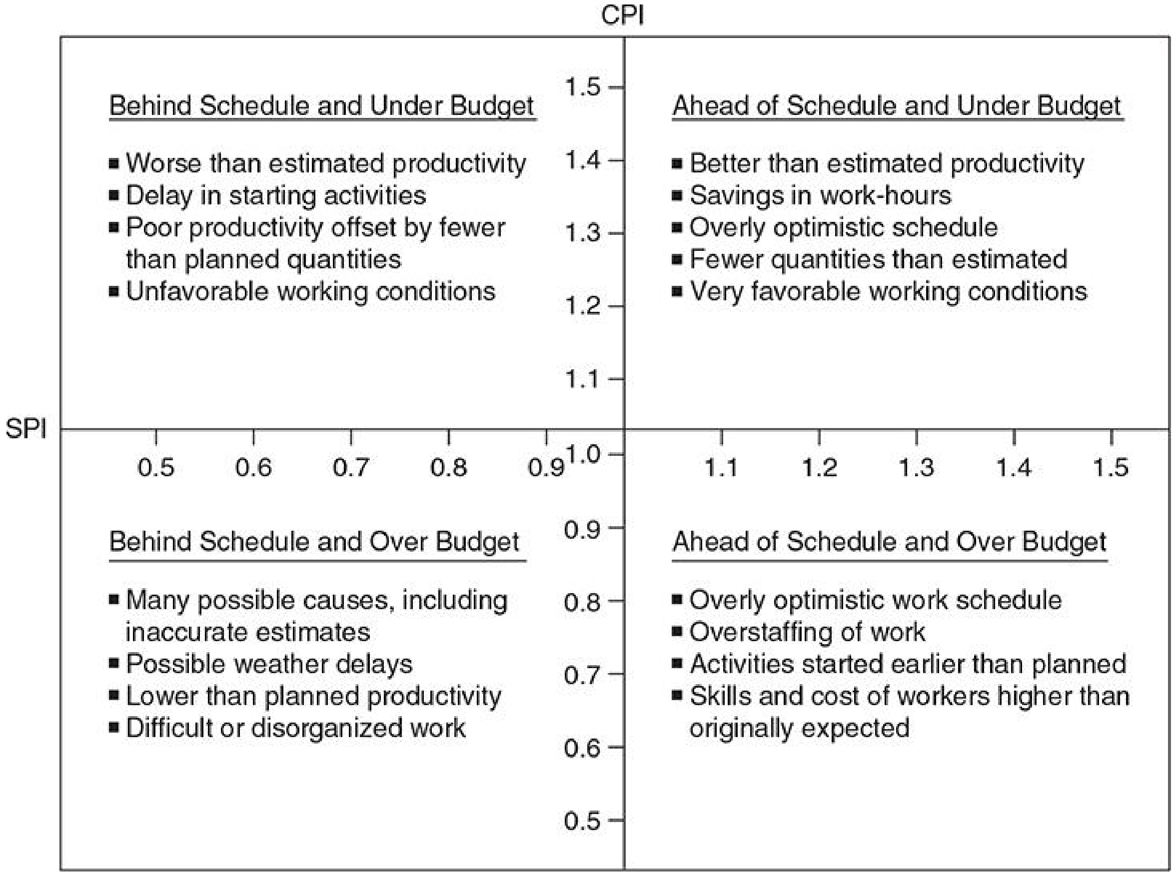


FIGURE 13-17 Various interpretations of CPI and SPI.

图 13-17 对 CPI 和 SPI 的各种解读。

If **CPI** is *greater* than 1.0:

Overestimated amount of effort required.

Measurement of percent complete is too optimistic.

Better than planned productivity.

If **CPI** is *less* than 1.0:

Underestimated amount of effort required.

Measurement of percent complete is too pessimistic.

Poorer than planned productivity.

If **SPI** is *greater* than 1.0:

Started work prior to or out of sequence to logical constraints.

Better than planned productivity resulting in activities completing early.

Working on critical activities.

If **SPI** is *less* than 1.0:

Understaffing or underequipping activities.

Delayed in starting of activities.

Work is more difficult than planned.

Improper sequencing of activities.

Work is disorganized.